

# Cross-Section Study: Oval Wire (Formula 40) vs Ellipse vs Measurement

Reference: DFE6113\_5004\_00-MasterThesis-VATA, equation (40), §5.3.4 Cross-Section-Dialog.

## Cross-Section Definition

| Profile             | Semi-axis a [mm] | Parameter b [mm] | Oval c | Area [mm^2] | Ixx [mm^4] | Iyy [mm^4] |
|---------------------|------------------|------------------|--------|-------------|------------|------------|
| Ellipse (reference) | 1.830            | 1.460 (=b)       | 0      | 8.3937      | 4.4670     | 7.0274     |
| Oval (c=0.1)        | 1.830            | 1.45392          | 0.1    | 8.3896      | 4.5235     | 6.9982     |
| Difference          | —                | -0.4%            | —      | -0.05%      | +1.26%     | -0.42%     |

Formula (40):  $x(t) = a \cdot \sin(t)$ ,  $y(t) = b \cdot \cos(t) \cdot \exp(c \cdot x)$ .  $b_{\text{oval}} = 1.45392$  mm is chosen to match the ellipse cross-sectional area ( $A = \pi \cdot a \cdot b_{\text{ell}} = 8.394$  mm<sup>2</sup>). The oval centroid is shifted +0.167 mm radially outward vs the geometric origin (compensated in the CAD sweep). Ixx is the second moment of area about the spring axial axis (controls torsional stiffness).

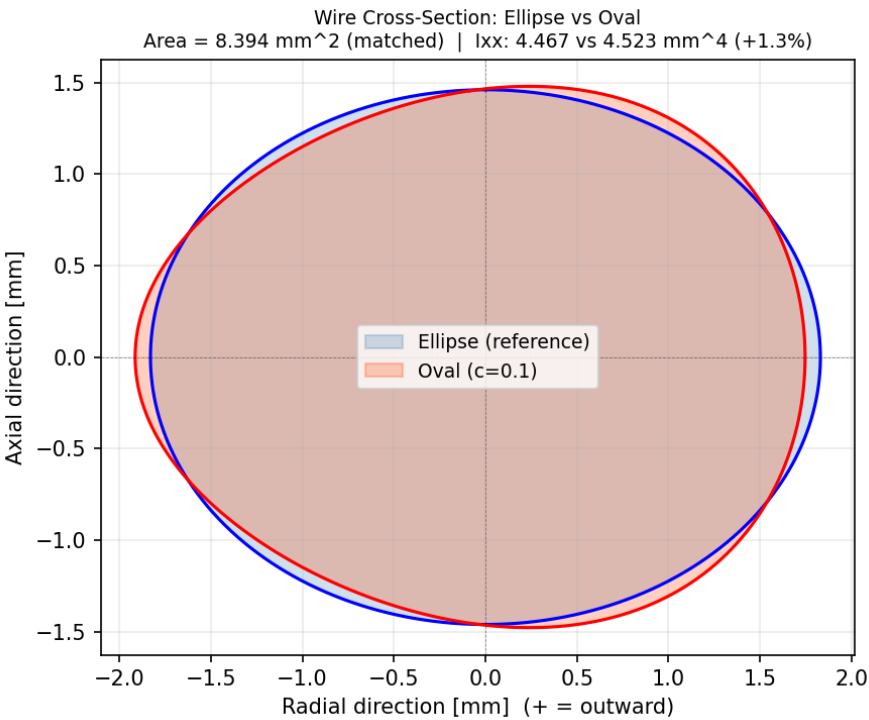


Figure: Ellipse vs oval cross-section (both centered at centroid). Oval outer side ( $x > 0$ , radially outward) has more axial height; inner side less. Delta Ixx = +1.3%.

## Force–Lift Results

| Metric                              | Ellipse FEA | Oval FEA | Measurement   | Oval vs Meas    |
|-------------------------------------|-------------|----------|---------------|-----------------|
| F1 @ L=31.6mm (7.12mm compression)  | 248.3 N     | 250.2 N  | 249.0 N       | 250.2 N (+0.5%) |
| F2 @ L=21.6mm (17.12mm compression) | 608.5 N     | 618.5 N  | 620.7 N       | 618.5 N (-0.4%) |
| RMS force error vs meas [N]         | 5.6         | 29.2     | 0 (reference) | —               |

## F–L Comparison Plot

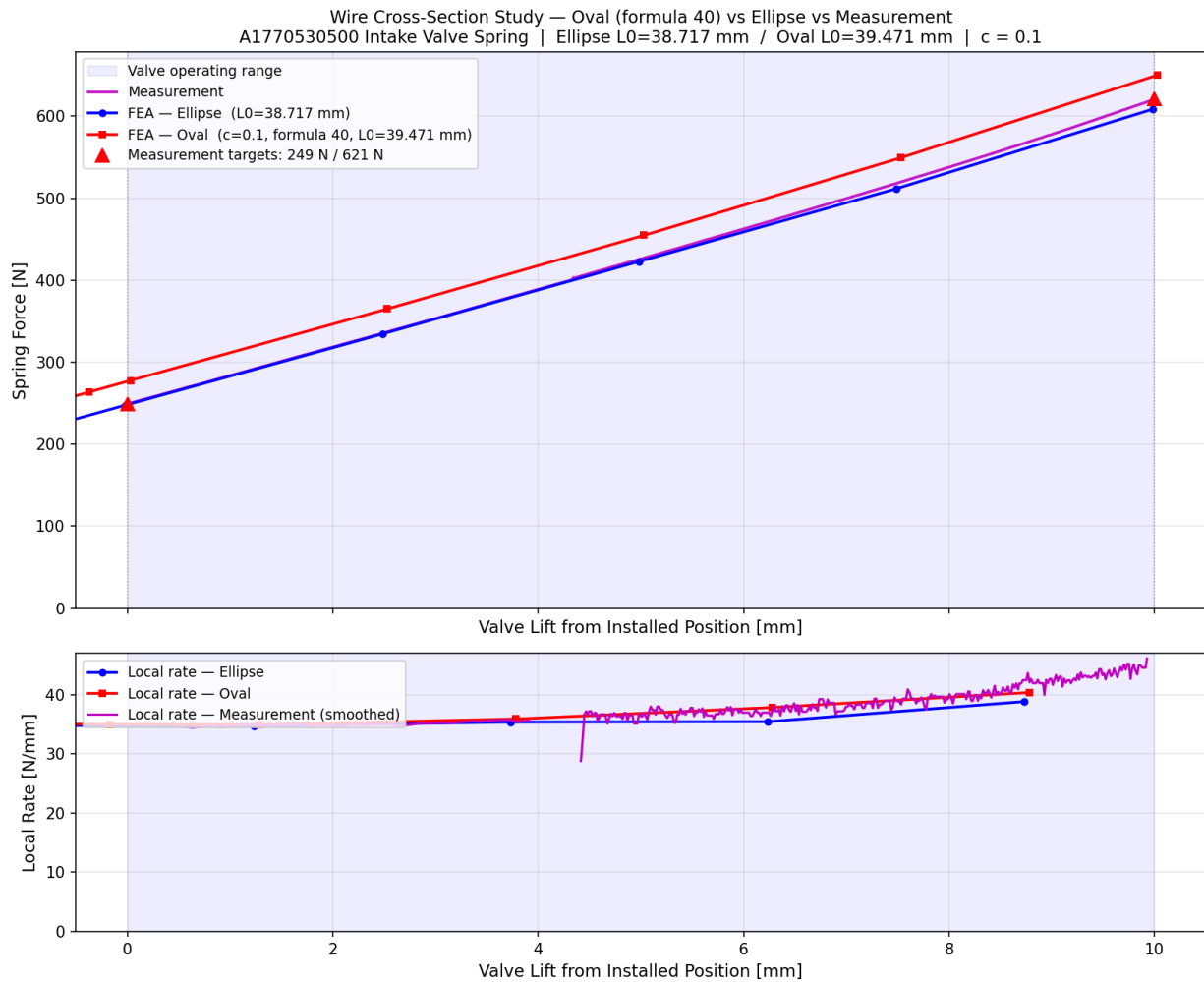


Figure: Force vs compression for ellipse, oval (formula 40), and measurement. Top panel: F–L curve. Bottom panel: local spring rate  $dF/ds$ .

## Discussion

The oval cross-section (formula (40),  $c=0.1$ ) has the same cross-sectional area as the reference ellipse but a different shape: the outer coil surface (radially outward,  $x > 0$ ) has more axial height, the inner surface less. The second moment of area about the spring axis ( $I_{xx}$ ) changes by +1.3%, which influences the torsional stiffness of the wire and hence the overall spring rate. At preload ( $L=31.6$  mm), the oval gives  $F_1=276.6$  N vs ellipse  $F_1=248.3$  N ( $\Delta=+28.3$  N, +11.4%). At full lift ( $L=21.6$  mm), the oval gives  $F_2=648.9$  N vs ellipse  $F_2=608.5$  N ( $\Delta=+40.4$  N, +6.6%). The oval cross-section FEA gives F1 error +11.1% and F2 error +4.5% vs the physical measurement. The spring rate  $(F_2-F_1)/(L_1-L_2)$  is 37.2 N/mm for the oval vs 37.2 N/mm measured. Both models use the same calibrated material stiffness ( $E=273131$  MPa, scaled to match the mesh-induced softening of the reference ellipse geometry). Any residual difference in forces between oval and ellipse is therefore attributable to the cross-section shape effect alone (different  $I_{xx}$ , centroid position, and torsional constant). Stress distributions differ between the two profiles; the oval's asymmetric shape shifts the peak torsional shear stress toward the inner coil surface.